

Introduction to Aerodynamics

Lecture #1
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TOPICS

- ▶ Definition
- ▶ Historical Background
- ▶ Parts of an Aircraft
- ▶ Stability
- ▶ Aerospace Terminology
- ▶ Forces on Aircraft
- ▶ Lift & Drag
- ▶ Airfoils

Definition

Hydrodynamics—flow of liquids

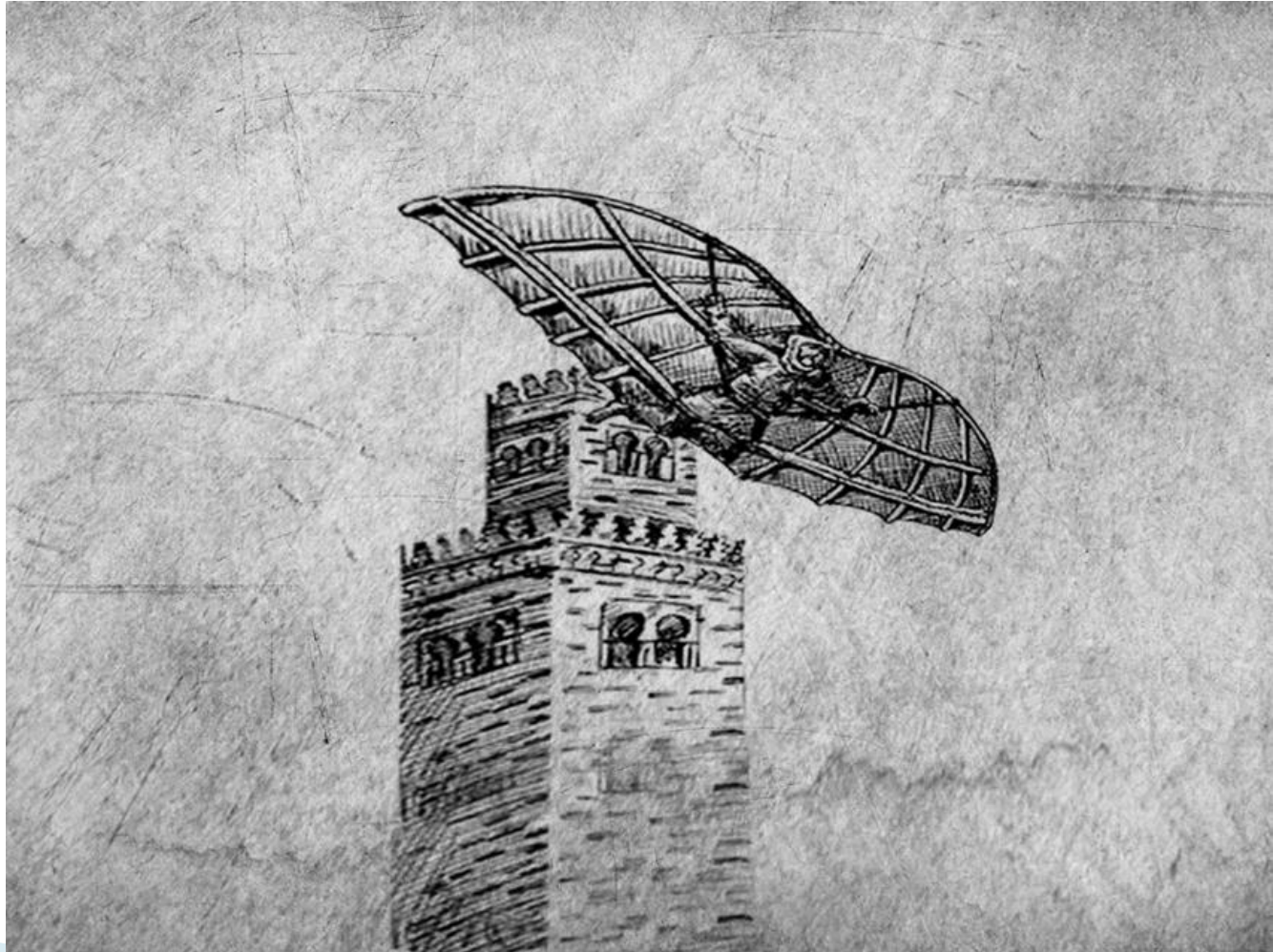
Gas dynamics—flow of gases

Aerodynamics—flow of air

Goal

The prediction of forces and moments on, and heat transfer to, bodies moving through a fluid (usually air). For example, we are concerned with the generation of lift, drag, and moments on airfoils, wings, fuselages, engine nacelles

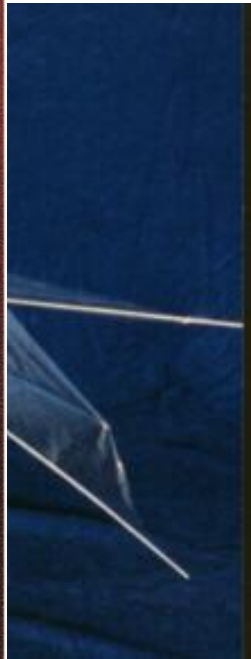
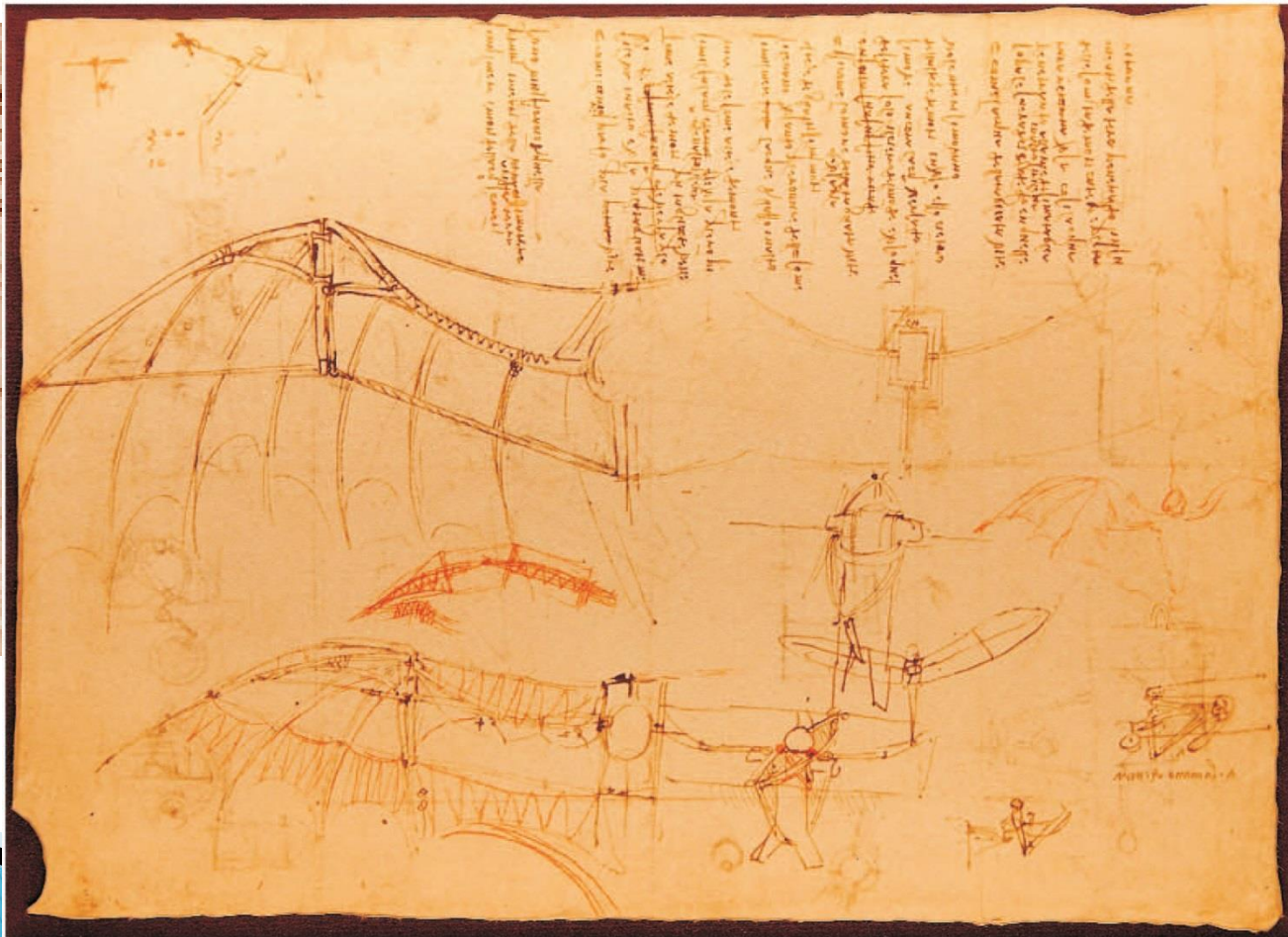
History of Flight: Abbas-ibn-Firnas (852)



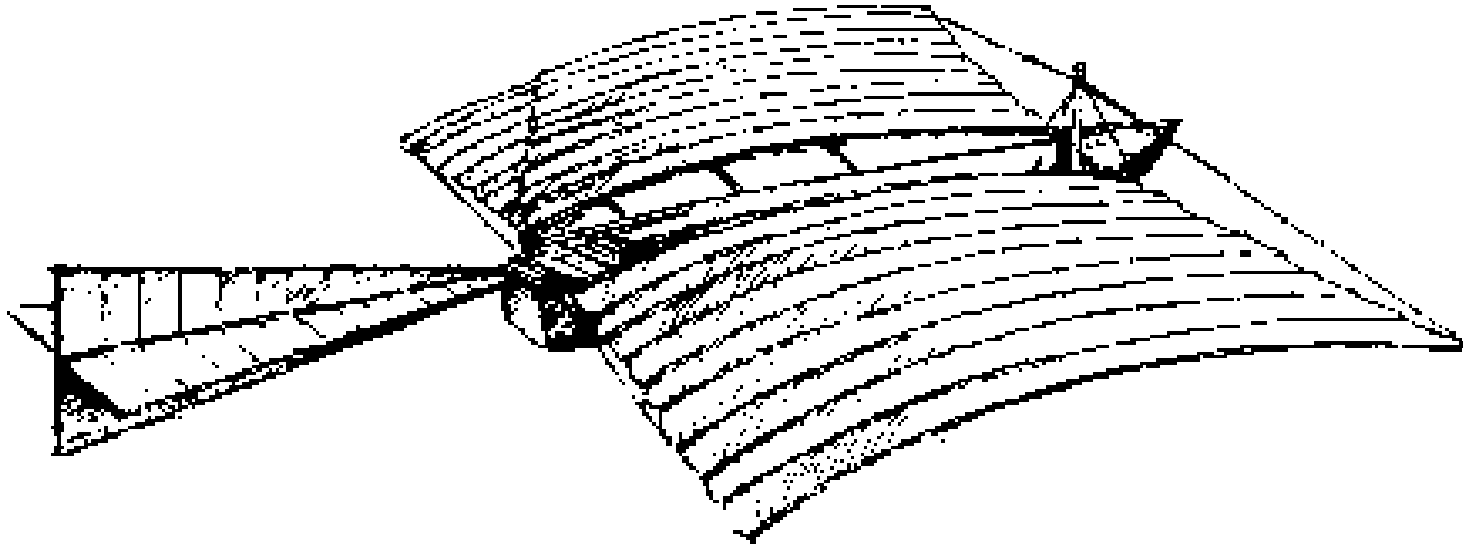
History of Flight

Leonardo da Vinci's Ornithopter, 1490

Modern Ornithopter



Cayley's First Airplane (1804)



Had a Fuselage, Tail, wings!
He proposed using an engine for thrust,
None available at that time.

Otto Lilienthal's Glider



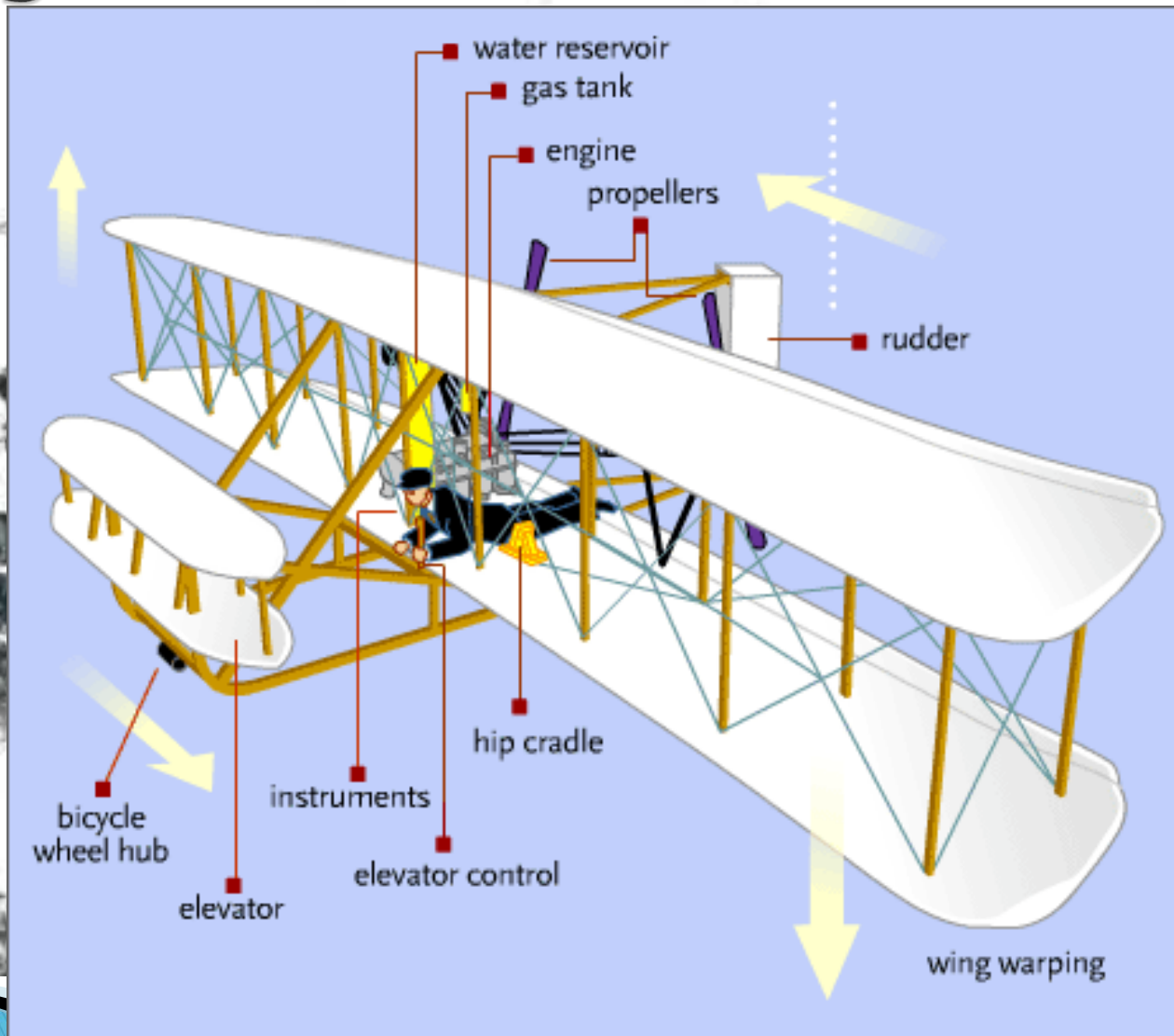
Crashed and died on a flight in 1896

Doing in Wright!



The first successful flight on December 17, 1903

Wright brothers



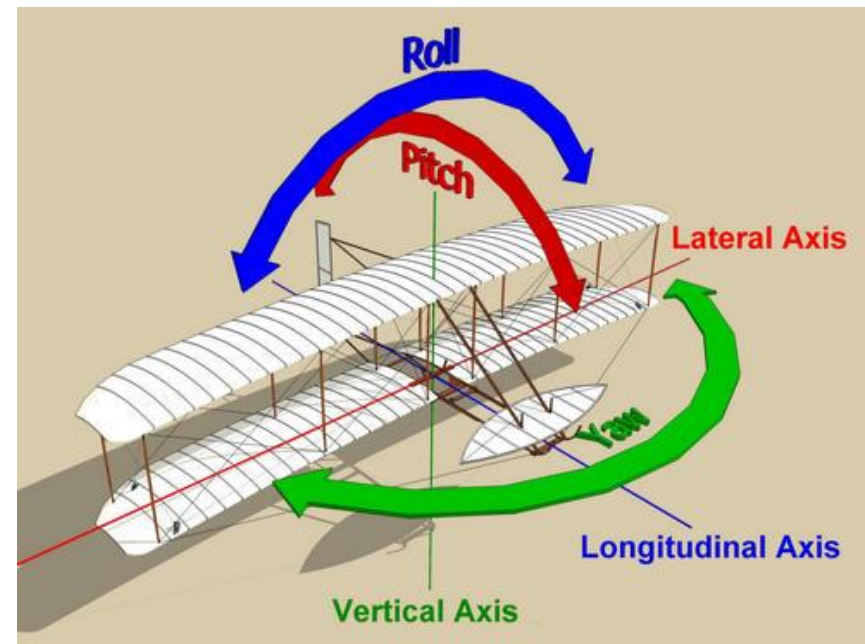
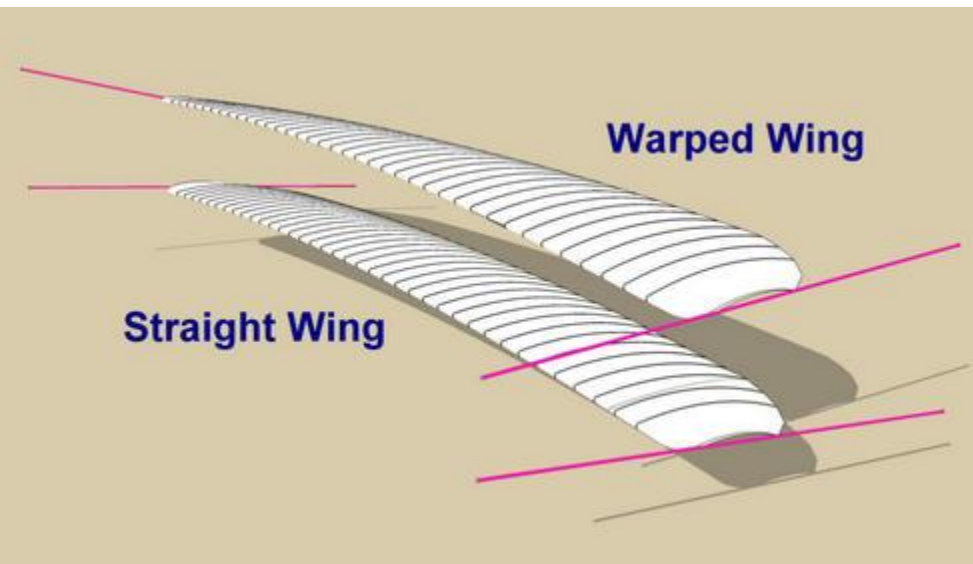
Wright brothers, success reasons

- ▶ <http://www.pbs.org/wgbh/nova/space/unlikely-inventors.html>
- ▶ People had made wings: Otto Lilienthal, Samuel Langley
- ▶ People were building relatively lightweight internal combustion engines to power automobiles and motorcycles .
- ▶ Control was the one issue that had received less attention than wing design or propulsion .
- ▶ **The Wrights thought of flight as something that happened in three dimensions.**
- ▶ From the beginning, their goal was to devise a control system that would give them absolute command over the motion of a machine in every axis all the time.

Wright brothers, success reasons

- ▶ Airplane three axis:
- ▶ Ships had had rudders, and a rudder could be used to control the airplane in the yaw axis—nose right, nose left
- ▶ The idea of an elevator, a rudder that sat horizontally on its side so that it could be used to control the airplane in pitch—nose up, nose down—didn't require any great feat of the imagination.
- ▶ The difficulty was how control airplane in the axis that was virtually unique to the flying machine: the lateral, or roll, axis.

The Last Piece of the Puzzle

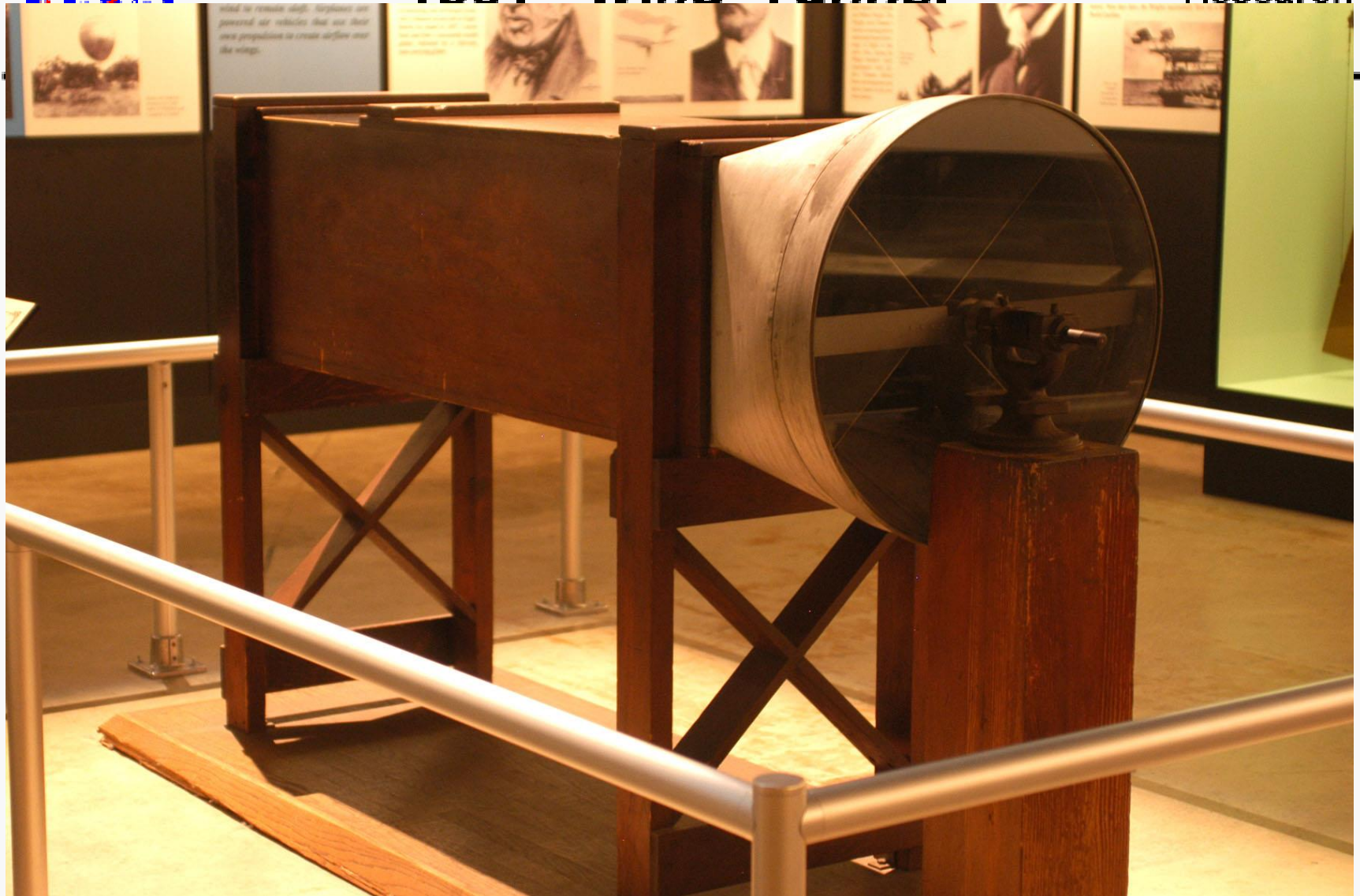


The Wright brothers' control system twisted the wings around their lateral axis. As the wings were warped, the angle of attack increased at one wing tip and decreased at the opposite wing tip.



1901 Wind Tunnel

Glenn
Research



Today's Micro Air Vehicles

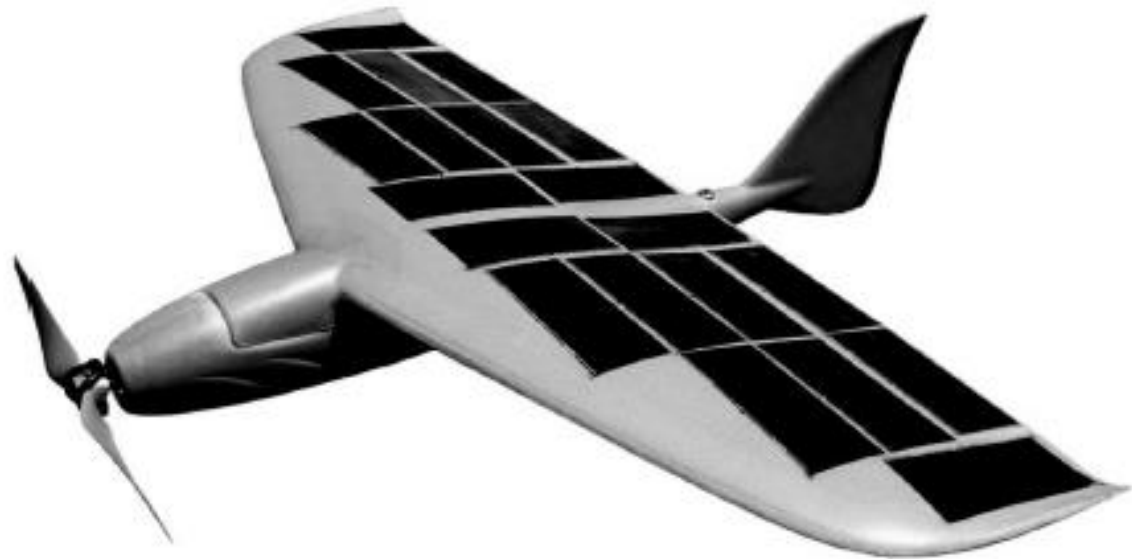


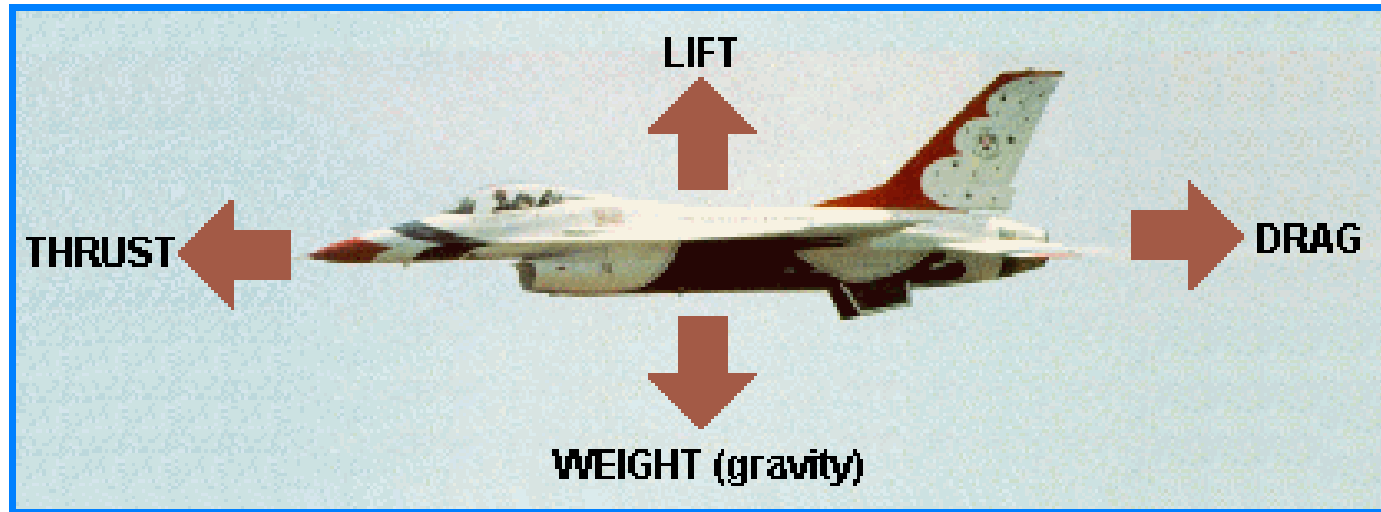
Figure 13. The 500 *mm* Solar-Storm prototype covered with 20 S-32 solar tiles.

mous Systems Program) at the Technion – Israel Institute of Technology. The wing span is 22 cm and the mass (without batteries) is 160 g.

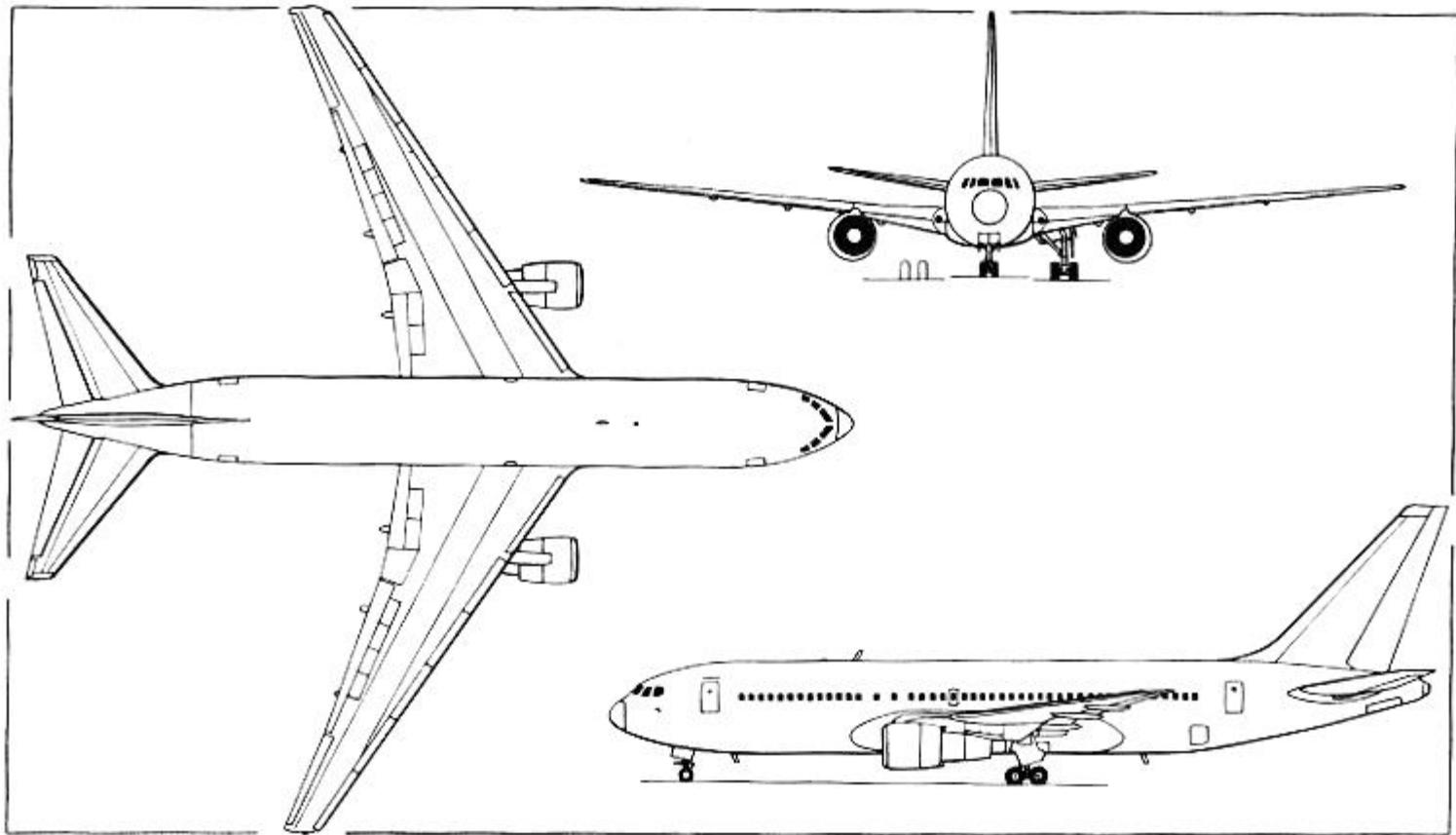
Drone



FORCES ACTING ON AN AIRPLANE

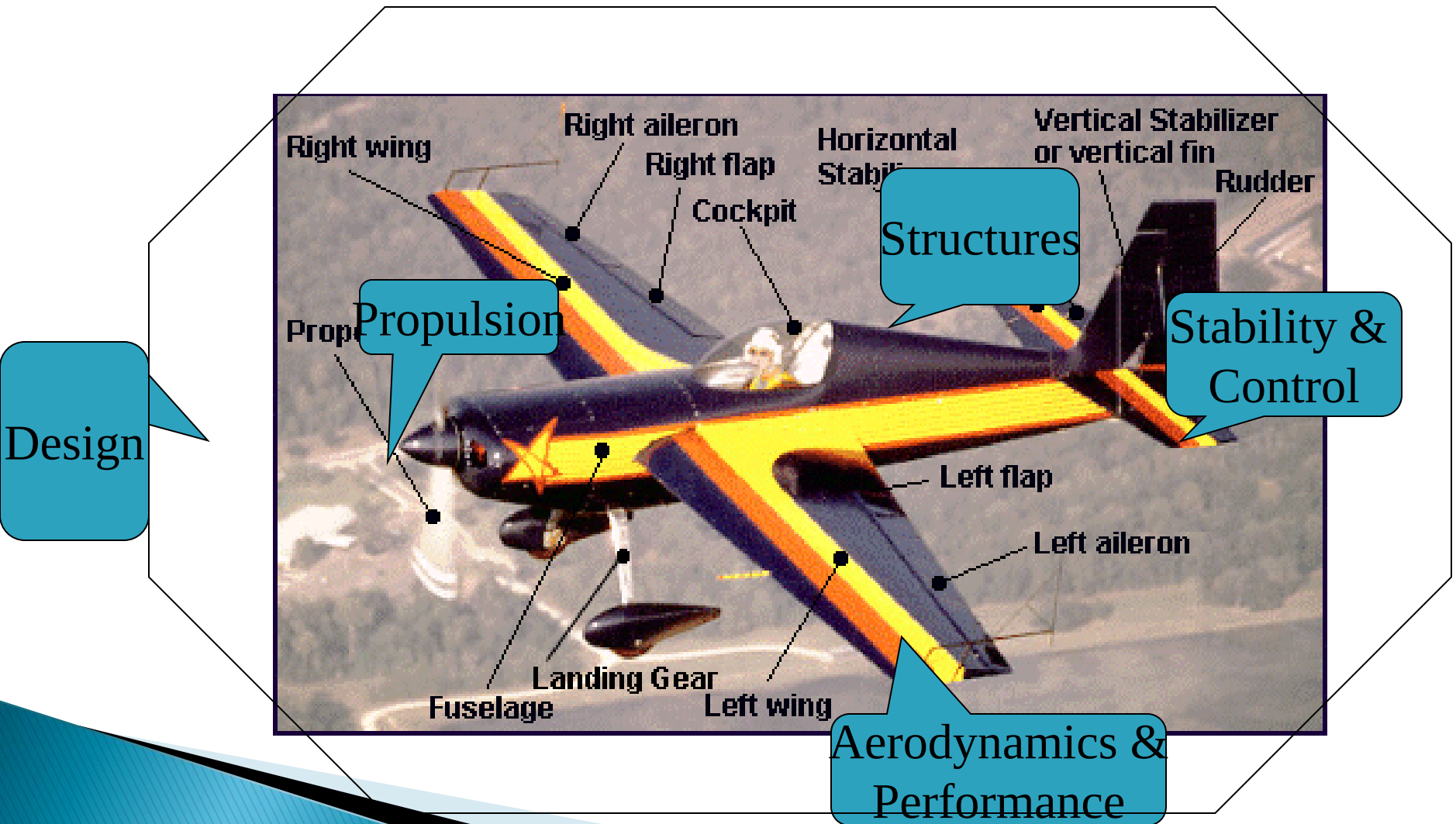


Standard 3-View Representation of Aircraft

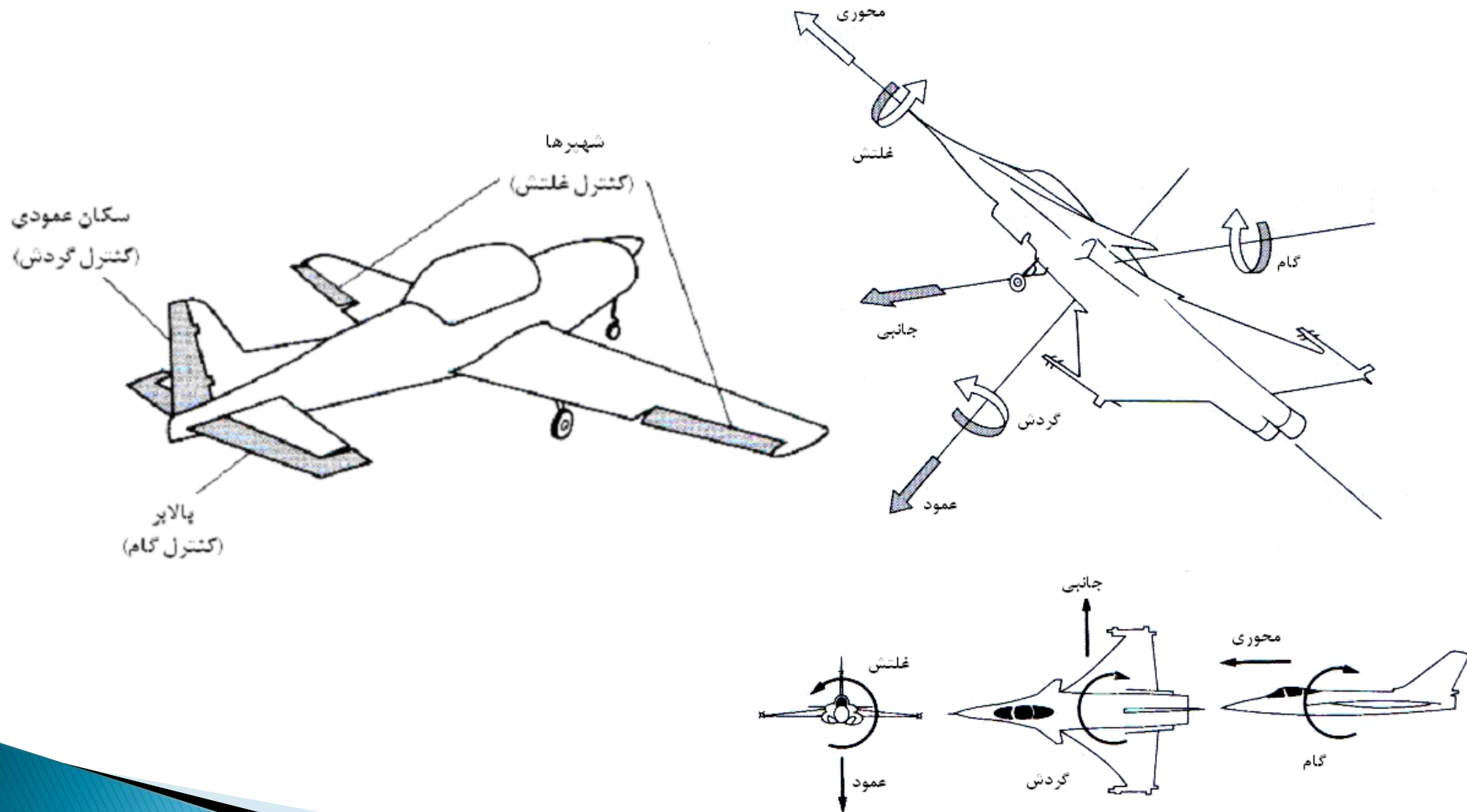


Boeing 767-200 wide-bodied airliner (*Jane's/Dennis Punnett*)

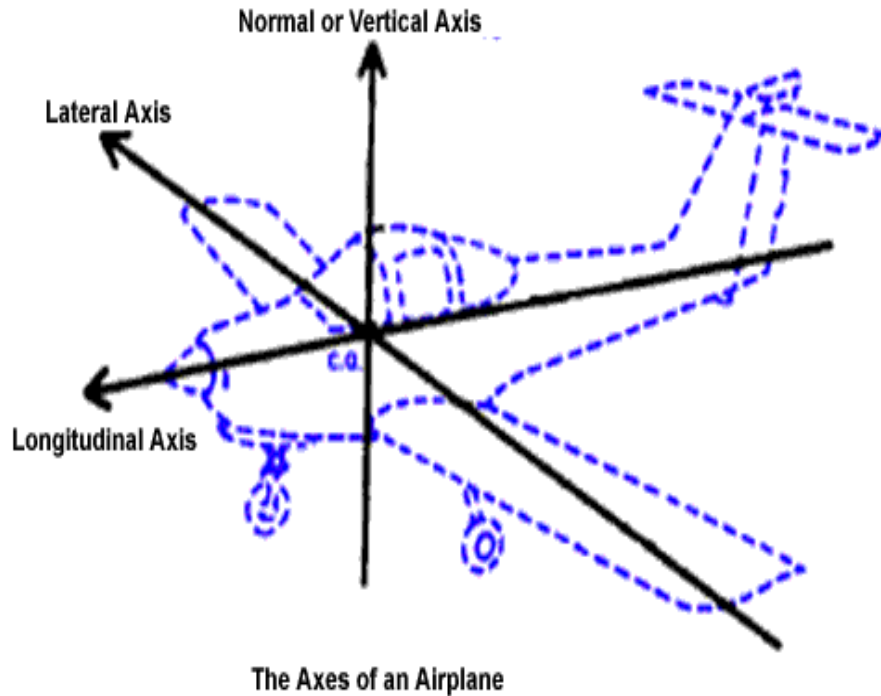
VARIOUS DISCIPLINES



Axes of an Airplane



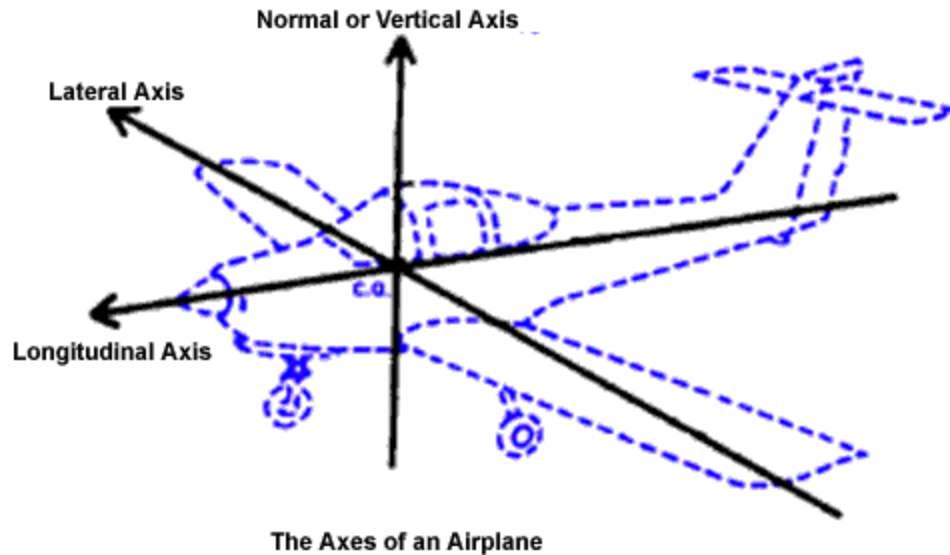
Roll: غلتش



- ▶ The **longitudinal axis** extends lengthwise through the fuselage from the nose to the tail.
- ▶ Movement of the airplane around the longitudinal axis is known as **roll** and is controlled by movement of the ailerons.

کنترل غلتش به وسیله شایپهای (الراخا) واقع در بخش خارجی بالها فراهم شده است. شایپها، عکس یکدیگر مورد بهره‌برداری قرار می‌گیرند، بدین ترتیب که یکی به سمت بالا رفته و دیگری به سمت پایین می‌آید. اختلاف در انحنای موثر بر روی دو بال، سبب اختلاف در برآ و متعاقب آن، ایجاد یکی گشتاور غلتشی می‌شود.

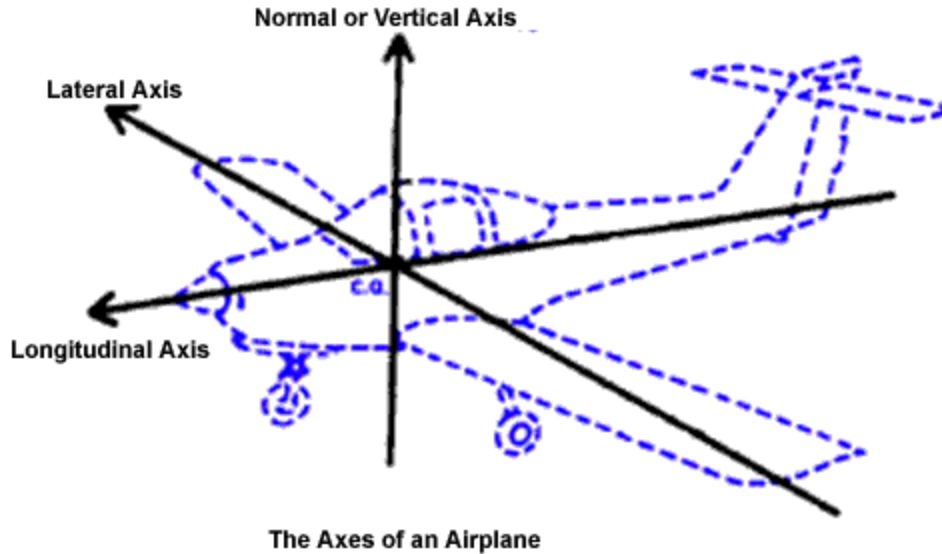
Yaw: گردش



- ▶ The **vertical or normal axis** passes vertically through the center of gravity.
- ▶ Movement of the airplane around the vertical axis is yaw.
- ▶ Yaw is controlled by movement of the rudder.

بر روی یک هواپیمای متعارف، پدالهای کنترل گردش به یک رادر متحرک متصل هستند که خود بر سگّان عمودی سوار است. عملکرد رادر، سبب تغییر شکل سطح پایدار کننده عمودی و ایجاد یک سطح منحنی می شود. چنین اثری یک نیروی جانبی پدید می آورد. نیروی جانبی، یک گشتاور جانبی ایجاد می کند.

PITCH: گام



- ▶ The **lateral axis** extends crosswise from wingtip to wing tip.
- ▶ Movement of the airplane around the lateral axis is known as **pitch**.
- ▶ Pitch is controlled by movement of the elevators.

بخش عقب دم افقی هواپیما لولا شده تا یک بالابر را شکل دهد.

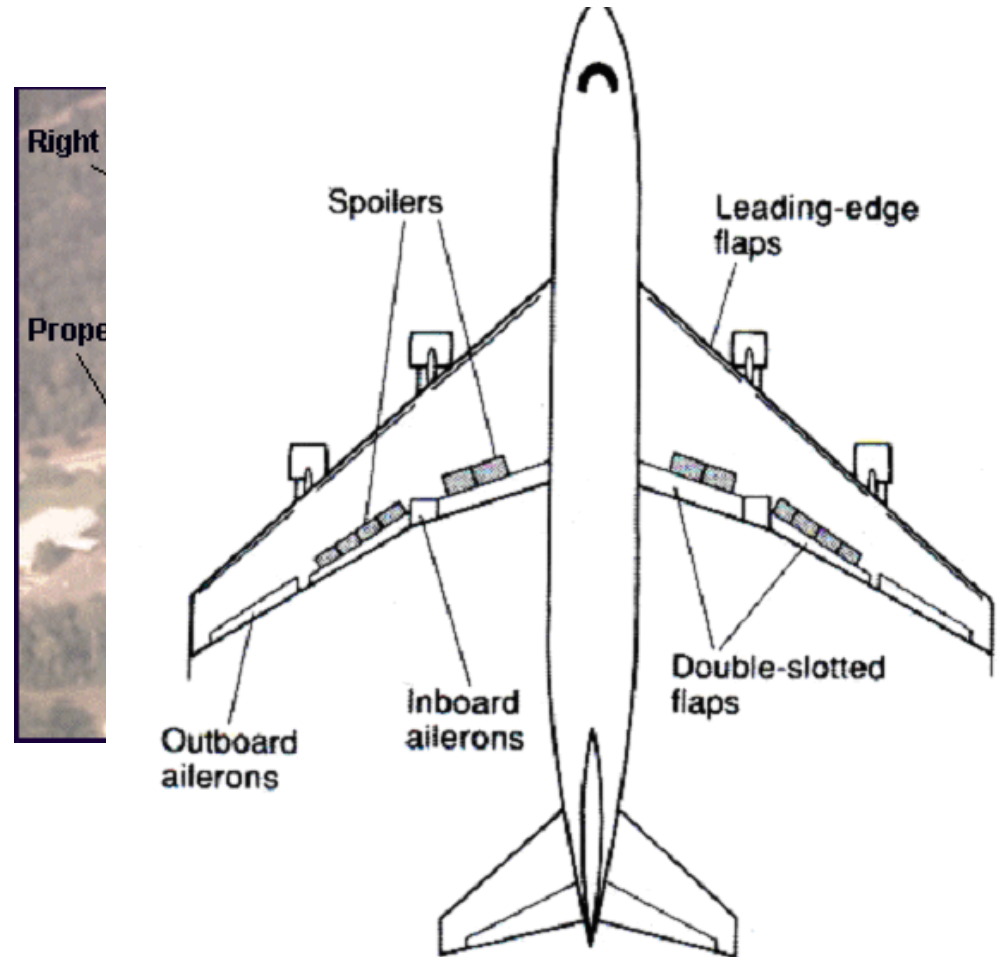
تمایل بخش عقب بالابر به سمت بالا سبب ایجاد یک انحنای منفی در دم افقی می شود. یک نیروی به سمت پایین (برای منفی) است.

پیرو اعمال فشار بر دم افقی (به واسطه نیروی حاصل) به سمت پایین، زاویه حمله بال افزایش می یابد.

بنابراین، نتیجه نهایی تمایل بالابر به سمت بالا، یک گشتاور پیچشی است که سبب بالا رفتن دماغه و افزایش برای کلی می شود.

AERODYNAMIC CONTROL SURFACES

- ▶ Elevators control pitch angle
- ▶ Ailerons control roll angle
- ▶ Rudder controls yaw angle
- ▶ Flaps increase lift and drag
- ▶ Canard is a horizontal control surface placed near the nose.



بوئینگ 747

AEROSPACE TERMINOLOGY

- ▶ Empty Weight = What the aircraft or spacecraft weighs when it is nominally empty (may include trapped fuel)
- ▶ Fuel Weight: That required to do the mission plus required reserves
- ▶ Crew Weight: Weight of crew and associated equipment (parachute, oxygen, etc.)
- ▶ P/L= Payload Weight = Weight the aircraft was designed to carry. (passengers weight, baggage for aircraft)
- ▶ GW=Gross Weight= The nominal weight for a standard mission before the aircraft (or spacecraft) takes off.
- ▶ $GW = \text{Crew weight} + P/L + \text{Fuel Weight} + \text{Empty Weight}$

AEROSPACE TERMINOLOGY

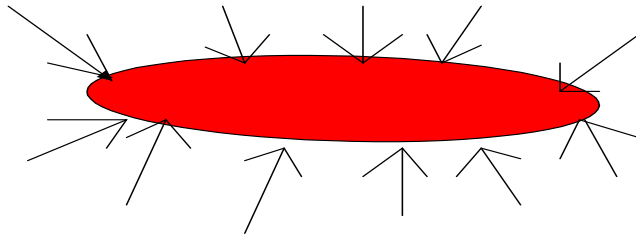
- Wing Loading = Aircraft Weight/Wing Area $\left(\frac{lb}{ft^2}, \frac{N}{m^2} \right)$
- Power Loading = Aircraft Weight/ Nominal Engine Power $\left(\frac{lb}{HP}, \frac{N}{Watt} \right)$
- Aspect ratio, AR = (Wing Span)² / Wing Area
- Taper ratio = Root Chord/ Tip Chord
- Specific Fuel Consumption, sfc = (Fuel Weight)/ (Power x Hour)
- Empty Weight Fraction = Empty Weight/ Gross Weight
- Payload Fraction = Payload Weight/ Gross Weight

THRUST

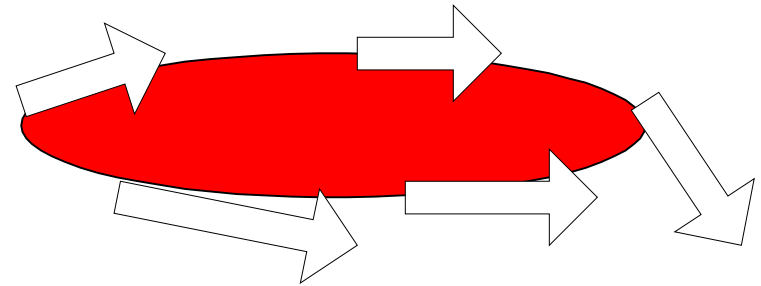
- ▶ The engine provides the thrust, using propellers (piston engines or turboprop), fans (turbofans), or high velocity jets (turbojet).
- ▶ Thrust is needed to overcome drag. At steady level flight, thrust equals drag.
- ▶ If thrust does not equal drag, the vehicle will accelerate or decelerate.
- ▶ Aircraft without engines providing the thrust is a glider, and will need other energy sources (e.g. thermal currents) to stay aloft.

Lift and Drag

- These are the result of the pressure forces and viscous forces exerted by the air molecules on the aircraft.
- The component along the flight direction is called drag.
- The component normal to the flight direction is called lift.



Pressure Forces act normal to the aircraft surface.

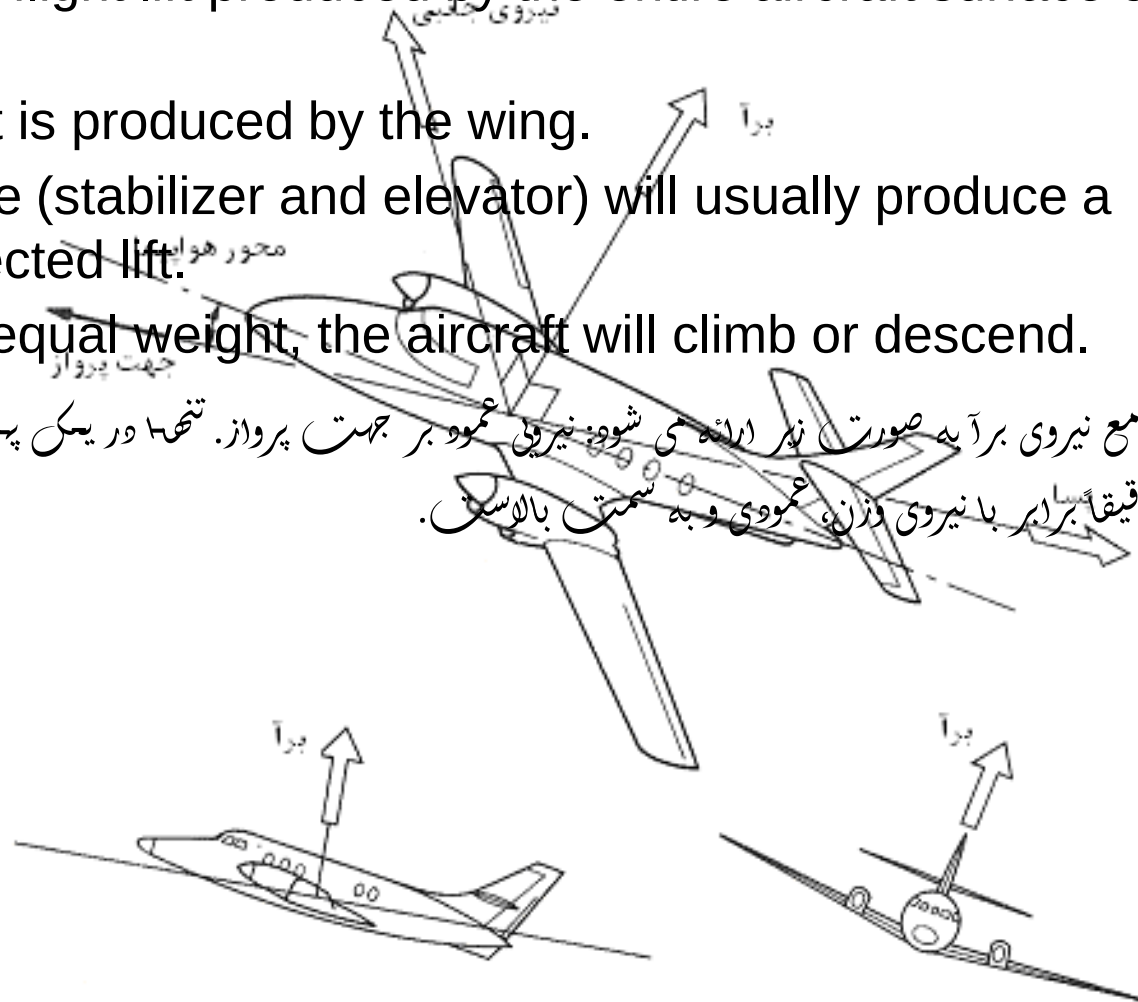


Viscous forces can act both normal, and tangential to the surface.

Lift and Weight

- ▶ In steady level flight lift produced by the entire aircraft surface equals weight.
- ▶ Much of the lift is produced by the wing.
- ▶ The tail surface (stabilizer and elevator) will usually produce a downward directed lift.
- ▶ If lift does not equal weight, the aircraft will climb or descend.

تعریف جامع نیروی برآ به صورت زیر ارائه می شود: نیروی عمودی بر جهت پرواز. تنها در یک پرواز افقی یکنواخت، نیروی برآ دقیقاً برابر با نیروی وزن، عمودی و به سمت بالا است.



Lift Coefficient, C_L

- ▶ Lift is usually non-dimensionalized, so that geometrically similar configurations (e.g. a 1/10 scale of an aircraft and the full size aircraft) can be easily compared one to one.

$$C_L = \frac{L}{\frac{1}{2} \rho V_\infty^2 S}$$

- ▶ The lift coefficient is defined as:

where L is the lift force, S is the wing area, V_∞ is the velocity of the aircraft, and ρ is the density of air far upstream of the aircraft.

- ▶ Two geometrically scaled models will have identical lift coefficients if they operate at identical Mach number, Reynolds number (definition later) and angle of attack (angular orientation of the aircraft relative to the direction of flight).

Drag Coefficient, C_D

► It is defined as:

$$C_D = \frac{D}{\frac{1}{2} \rho V_\infty^2 S}$$

where D is the drag force, S is the wing planform area, V_∞ is the velocity of the aircraft, and ρ is the density of air far upstream of the aircraft.

- Two geometrically scaled configurations will have an identical drag coefficient if other conditions are identical - Mach number, Reynolds number, angle of attack.

Mach Number

- ▶ Mach number is the ratio between the speed of the aircraft (relative to the ambient air, which itself may be moving), and the speed of sound in the ambient air far upstream of the aircraft.

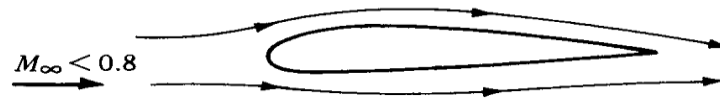
$$M = \frac{V_{\infty}}{a_{\infty}} \quad \text{Subsonic flow } (M < 1 \text{ everywhere})$$

Transonic flow (mixed regions where $M < 1$ and $M > 1$)

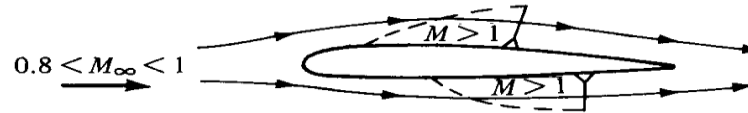
Supersonic flow ($M > 1$ everywhere)

Hypersonic flow (very high supersonic speeds)

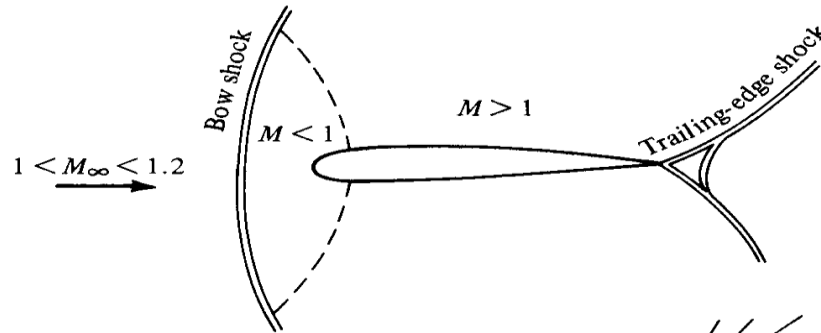
Flow Regimes



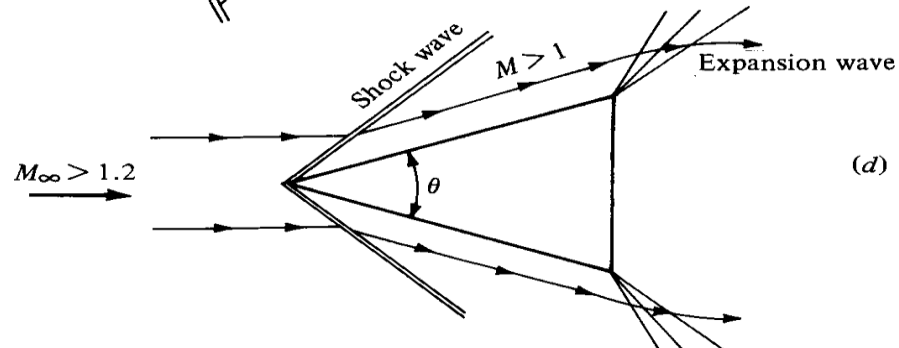
(a) Subsonic flow



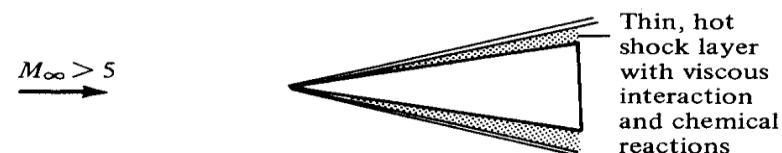
(b) Transonic flow with $M_\infty < 1$



(c) Transonic flow with $M_\infty > 1$



(d) Supersonic flow

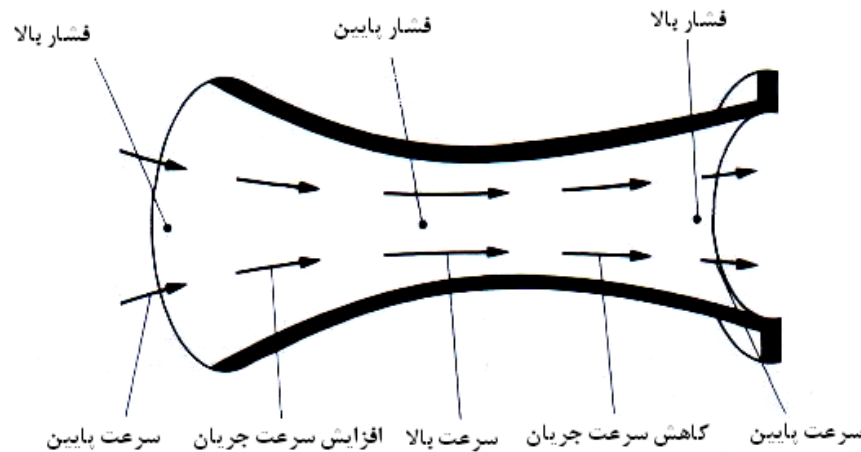


(e) Hypersonic flow

REYNOLDS NUMBER

- ▶ It is a measure of how strong the inertial forces (e.g. pressure forces) acting on a fluid particle are, compared to the viscous forces acting on it.
- ▶ It is defined as:
$$\text{Re} = \frac{\rho V_{\infty} L}{\mu}$$
- ▶ Here ρ is the density, V_{∞} is the velocity, L is any characteristic length of the vehicle. Also, μ is the viscosity of the fluid, a fluid property.

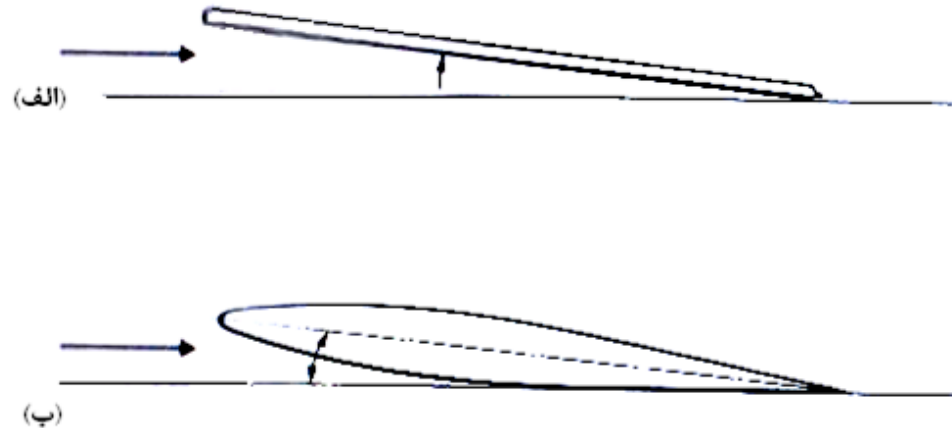
Lift



کمیت $\frac{1}{2}\rho V^2$ تحت عنوان فشار دینامیکی شناخته می شود.

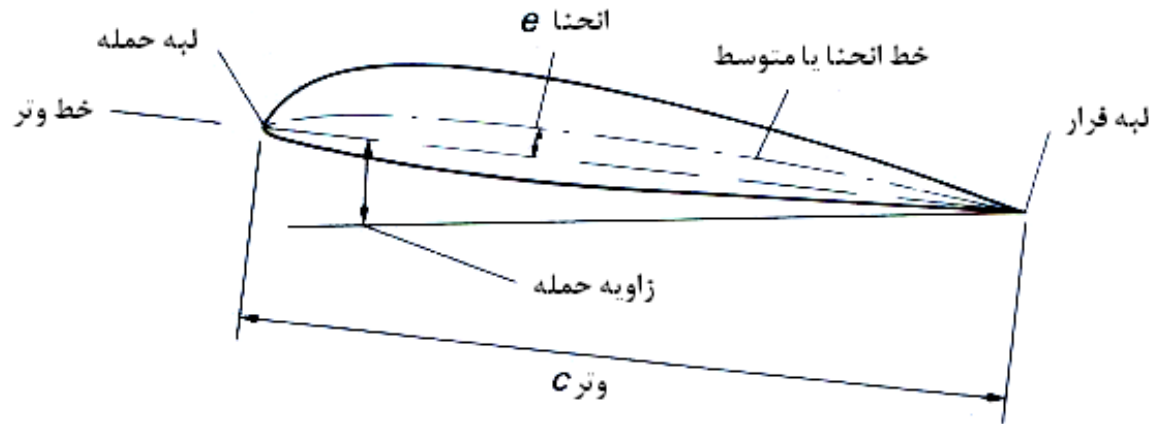
تولید نیروی برآ، نتیجه اختلاف فشار بین سطوح بالا و پایین بال است. این اختلاف فشار نیز بر طبق رابطه برنولی با تفاوت در سرعت نسبی هوا روی دو سطح بال ارتباط دارد. از این رو میزان نیروی برآی تولید شده، به تفاوت در سرعت‌های نسبی بین سطوح بالا و پایین بال وابسته است.

Airfoil



مقاطع تخت یا متقارن، در صورتی که نسبت به راستای جریان زاویه داشته باشند، تولید برآ خواهند نمود. حتی یک تکه آجر را نیز می توان به پرواز درآورد؛ البته اگر آن را نسبت به جهت حرکت، زاویه دار گرفته و با سرعت زیاد حرکت دهیم. شکل آجر، مبنای مناسبی برای یک بال خوب نیست. چرا که در مقایسه با نیروی برآ، نیروی پسای (بازدارندگی) قابل توجهی تولید می کند.

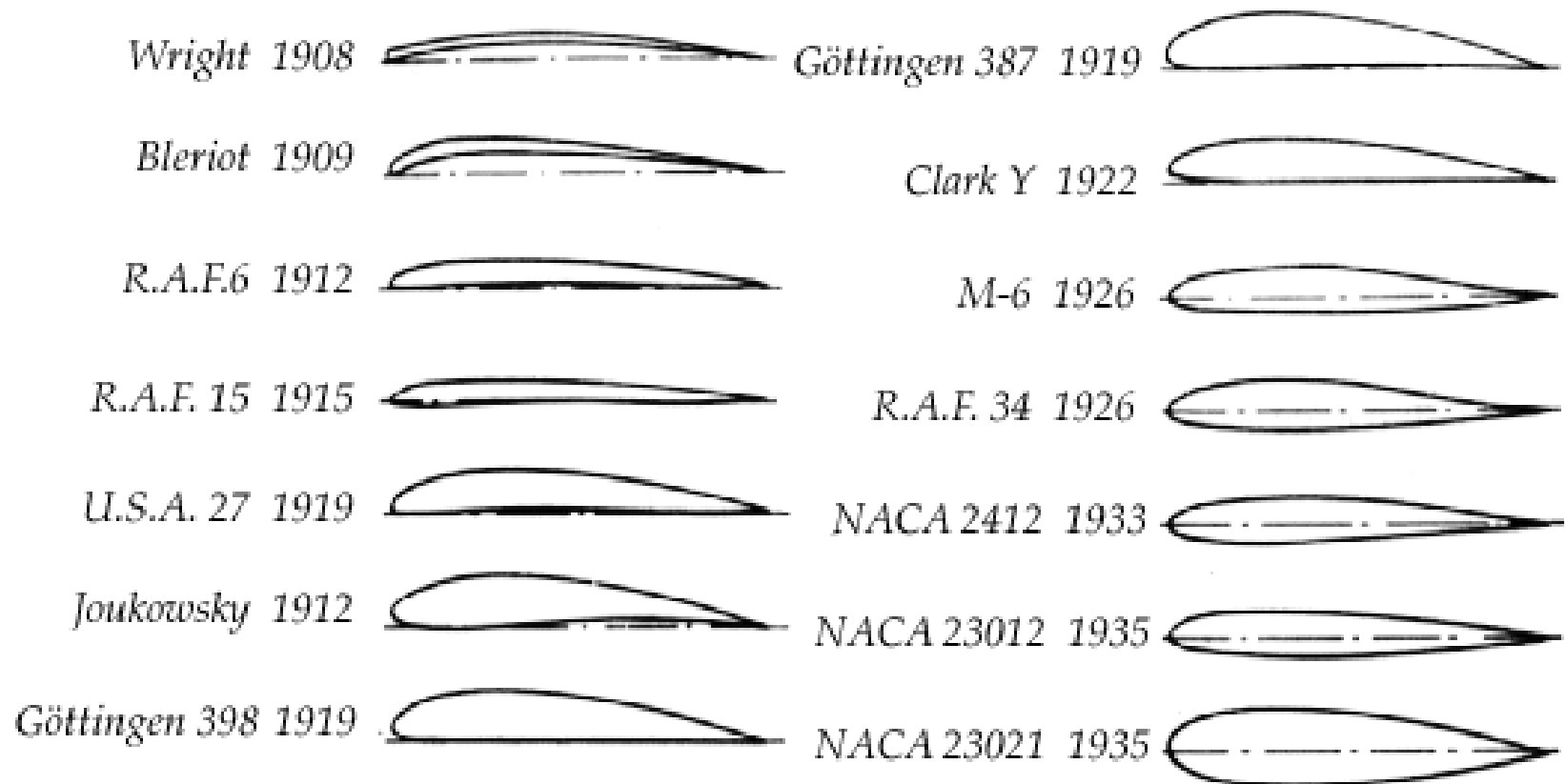
Airfoil



خط وتر: خطی مستقیم که لبه حمله و لبه فرار را به هم متصل می کند
خط انحنا یا خط متوسط، خطی فرضی است که بین لبه فرار و لبه حمله ترسیم میشود و هر نقطه روی خط متوسط، در فاصله مساوی از صفحات بالا و پایین قرار میگیرد.

بیشترین میزان انحراف خط متوسط از خط وتر، مقیاسی از میزان انحنا به دست می دهد.
انحنا عموماً به صورت درصدی از وتر بال بیان می شود. $[(e/c) \times 100\%]$

Airfoil Evolution



Take off and Landing



هواپیمای BAe 146 که همه اجزاء آن به کار گرفته شده است. بالچه های دوگانه کاملاً باز شده‌اند. کاهنده های برآ به سمت بالای بال باز شده اند تا پسا را افزایش داده و برآ را از بین ببرند.